

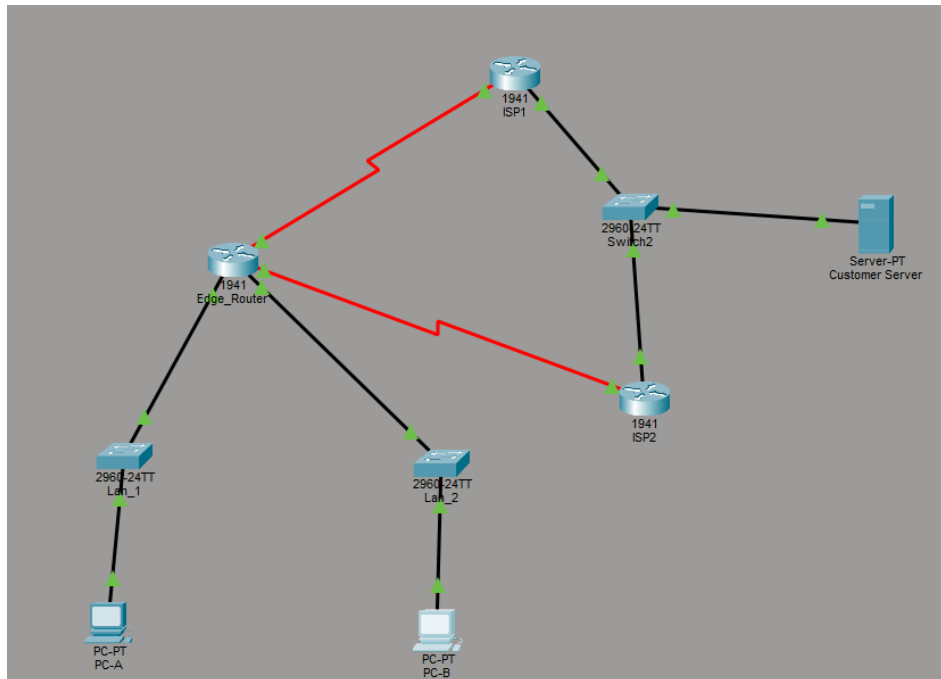
Lab 9

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ขออนุญาตทำโดยใช้ packet tracer ทั้งหมดเนื่องจากเวลาเรียนตรงกับวิชา GE ทำให้ไม่สามารถใช้ห้องLab ทำกิจกรรมได้

Addressing Table

Device	Interface	IP Address / Prefix
Edge_Router	S0/0/0	10.10.10.2/30
		2001:db8:a:1::2/64
	S0/0/1	10.10.10.6/30
		2001:db8:a:2::2/64
	G0/0	192.168.10.17/28
		2001:db8:1:10::1/64
ISP1	S0/0/0	10.10.10.1/30
		2001:db8:a:1::1/64
	G0/0	198.0.0.1/24
		2001:db8:f:f::1/64
ISP2	S0/0/1	10.10.10.5/30
		2001:db8:a:2::1/64
	G0/0	198.0.0.2/24
		2001:db8:f:f::2/64
PC-A	NIC	192.168.10.19/28
		2001:db8:1:10::19/64
PC-B	NIC	192.168.11.4/27
		2001:db8:1:11::45
Customer Server	NIC	198.0.0.10
		2001:db8:f:f::10



ตั้งค่า Edge_Router

```

Edge_Router
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Edge_Router
Edge_Router(config)#ipv6 unicast-routing
Edge_Router(config)#interface s0/0/0
Edge_Router(config-if)#ip address 10.10.10.2 255.255.255.252
Edge_Router(config-if)#ipv6 address 2001:db8:a:1::2/64
Edge_Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Edge_Router(config-if)#interface s0/0/1
Edge_Router(config-if)#ip address 10.10.10.6 255.255.255.252
Edge_Router(config-if)#ipv6 address 2001:db8:a:2::2/64
Edge_Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
Edge_Router(config-if)#interface g0/0
Edge_Router(config-if)#ip address 192.168.10.17 255.255.255.240
Edge_Router(config-if)#ipv6 address 2001:db8:1:10::1/64
Edge_Router(config-if)#no shutdown

Edge_Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

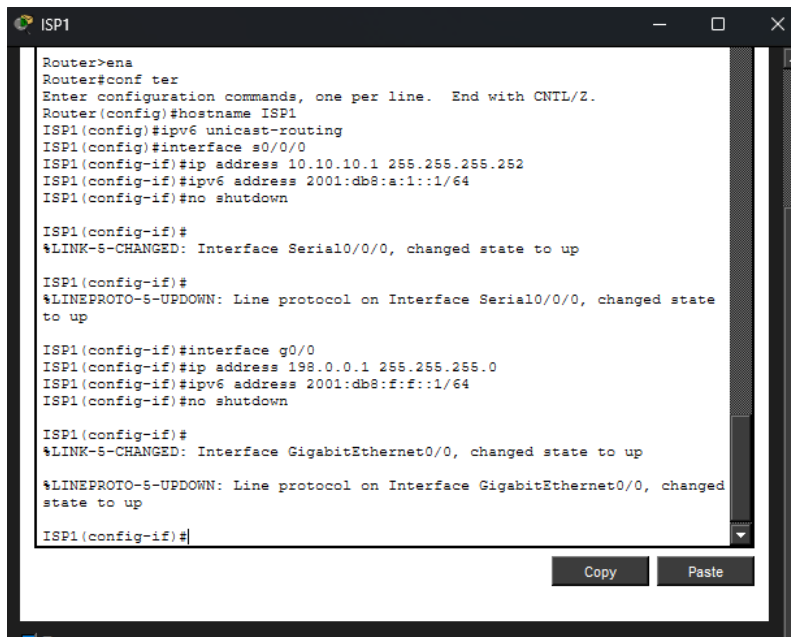
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up

Edge_Router(config-if)#interface g0/1
Edge_Router(config-if)#ip address 192.168.11.33 255.255.255.224
Edge_Router(config-if)#ipv6 address 2001:db8:1:11::1/64
  
```

ตั้งชื่อโฮสต์เป็น Edge_Router และเปิดใช้งาน ipv6 unicast-routing จากนั้นกำหนด IP Address ทั้งแบบ IPv4 และ IPv6 ให้กับอินเทอร์เฟซ S0/0/0 (เชื่อมต่อ ISP1), S0/0/1 (เชื่อมต่อ ISP2), G0/0 (เชื่อมต่อ LAN1) และ G0/1 (เชื่อมต่อ LAN2) พร้อม

เปิดใช้งานแต่ละพอร์ตด้วยคำสั่ง no shutdown โดยพอร์ต G0/0 แสดงสถานะเปลี่ยนเป็น up ยืนยันว่าการเชื่อมต่อทำงานได้ปกติ

ตั้งค่า ISP1



```
Router>ena
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP1
ISP1(config)#ipv6 unicast-routing
ISP1(config)#interface s0/0/0
ISP1(config-if)#ip address 10.10.10.1 255.255.255.252
ISP1(config-if)#ipv6 address 2001:db8:a::1/64
ISP1(config-if)#no shutdown

ISP1(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

ISP1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state
to up

ISP1(config-if)#interface g0/0
ISP1(config-if)#ip address 198.0.0.1 255.255.255.0
ISP1(config-if)#ipv6 address 2001:db8:f::1/64
ISP1(config-if)#no shutdown

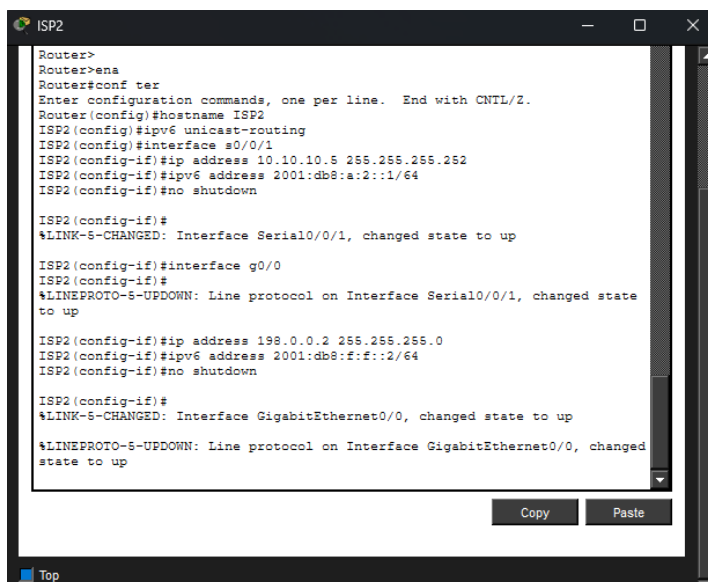
ISP1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed
state to up

ISP1(config-if)#
```

ตั้งชื่อโฮสต์เป็น ISP1 เปิดใช้งาน ipv6 unicast-routing และกำหนดค่า IP Address ทั้ง IPv4 และ IPv6 ให้กับอินเทอร์เฟซ S0/0/0 และ G0/0 ทั้งสองพอร์ตเปิดใช้งานด้วย no shutdown และแสดงสถานะขึ้นเป็น up เรียบร้อย

ตั้งค่า ISP2



```
ISP2
Router>
Router>ena
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP2
ISP2(config)#ipv6 unicast-routing
ISP2(config)#interface s0/0/1
ISP2(config-if)#ip address 10.10.10.5 255.255.255.252
ISP2(config-if)#ipv6 address 2001:db8:a::1/64
ISP2(config-if)#no shutdown
ISP2(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up
ISP2(config-if)#interface g0/0
ISP2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
ISP2(config-if)#ip address 198.0.0.2 255.255.255.0
ISP2(config-if)#ipv6 address 2001:db8:f::2/64
ISP2(config-if)#no shutdown
ISP2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

ตั้งชื่อโฮสต์เป็น ISP2 เปิดใช้งาน ipv6 unicast-routing และกำหนดค่า IP Address ทั้ง IPv4 และ IPv6 ให้กับอินเทอร์เฟซ S0/0/1 และ G0/0 โดยทำหน้าที่เป็นเส้นทางสำรอง ในกรณีที่เส้นทางหลักผ่าน ISP1 ใช้งานไม่ได้

Objectives

In this Packet Tracer summary activity, you will configure static, default, and floating static routes for both the IPv4 and IPv6 protocols.

- Configure IPv4 Static and Floating Static Default Routes.
- Configure IPv6 static and floating static default routes.
- Configure IPv4 static and floating static routes to internal LANs.
- Configure IPv6 static and floating static routes to the internal LANS.
- Configure IPv4 host routes.
- Configure IPv6 host routes.

Background / Scenario

In this activity, you will configure IPv4 and IPv6 default static and floating static routes.

Note: The static routing approach that is used in this lab is used to assess your ability to configure different types of static routes only. This approach may not reflect networking best practices.

Instructions

Part 1: Configure IPv4 Static and Floating Static Default Routes

The PT network requires static routes to provide internet access to the internal LAN users through the ISPs. In addition, the ISP routers require static routes to reach the internal LANs. In this part of the activity, you will configure an IPv4 static default route and a floating default route to add redundancy to the network.

Step 1: Configure an IPv4 static default route.

On Edge_Router, configure a **directly connected** IPv4 default static route. This primary default route should be through router **ISP1**.

Step 2: Configure an IPv4 floating static default route.

On Edge_Router, configure a **directly connected** IPv4 floating static default route. This default route should be through router **ISP2**. It should have an administrative distance of **5**.

Part 2: Configure IPv6 Static and Floating Static Default Routes

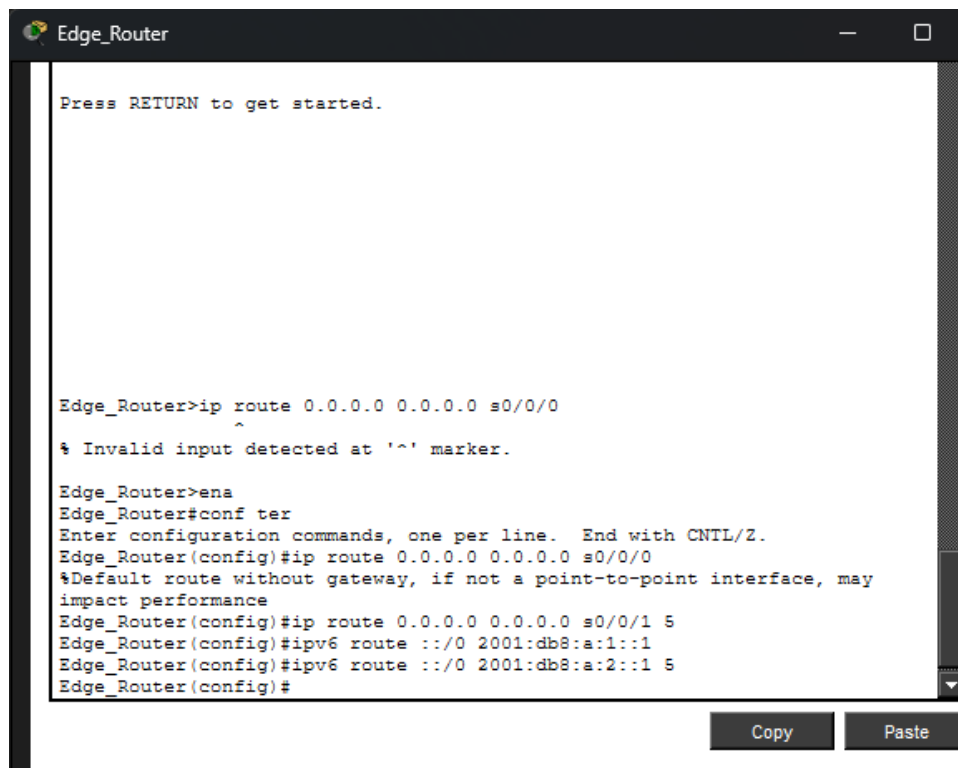
In this part of the activity, you will configure IPv6 static default and floating static default routes for IPv6.

Step 1: Configure an IPv6 static default route.

On Edge_Router, configure a **next hop** static default route. This primary default route should be through router **ISP1**.

Step 2: Configure an IPv6 floating static default route.

On Edge_Router, configure a **next hop** IPv6 floating static default route. The route should be via router **ISP2**. Use an administrative distance of **5**.

A screenshot of a terminal window titled 'Edge_Router'. The terminal shows a sequence of commands and their outputs. It starts with a prompt 'Press RETURN to get started.' followed by a series of configuration commands: 'ip route 0.0.0.0 0.0.0.0 s0/0/0', 'ena', 'conf ter', 'ip route 0.0.0.0 0.0.0.0 s0/0/0', 'ip route 0.0.0.0 0.0.0.0 s0/0/1 5', 'ipv6 route ::/0 2001:db8:a:1::1', and 'ipv6 route ::/0 2001:db8:a:2::1 5'. The terminal also shows some error messages like '% Invalid input detected at '^' marker.' and a warning about default routes without gateways. At the bottom of the terminal window, there are 'Copy' and 'Paste' buttons.

```
Edge_Router>ip route 0.0.0.0 0.0.0.0 s0/0/0
^
% Invalid input detected at '^' marker.

Edge_Router>ena
Edge_Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Edge_Router(config)#ip route 0.0.0.0 0.0.0.0 s0/0/0
%Default route without gateway, if not a point-to-point interface, may
impact performance
Edge_Router(config)#ip route 0.0.0.0 0.0.0.0 s0/0/1 5
Edge_Router(config)#ipv6 route ::/0 2001:db8:a:1::1
Edge_Router(config)#ipv6 route ::/0 2001:db8:a:2::1 5
Edge_Router(config)#
```

กำหนดเส้นทางเริ่มต้น แบบ IPv4 หลักผ่าน S0/0/0 ไปยัง ISP1 ด้วยค่า Administrative Distance ปกติ และกำหนด floating default route สำรองผ่าน S0/0/1 ไปยัง ISP2 โดยระบุค่า AD=5 เพื่อให้เส้นทางนี้ทำงานเฉพาะเมื่อเส้นทางหลักใช้งานไม่ได้ ในส่วนของ IPv6 ก็กำหนดในลักษณะเดียวกัน โดยใช้ next-hop address ของ ISP1 เป็นเส้นทางหลักและ ISP2 เป็นเส้นทางสำรอง

Part 3: Configure IPv4 Static and Floating Static Routes to the Internal LANs

In this part of the lab you will configure static and floating static routes from the ISP routers to the internal LANs.

Step 1: Configure IPv4 static routes to the internal LANs.

- On ISP1, configure a **next hop** IPv4 static route to the **LAN 1** network through Edge_Router.
- On ISP1, configure a **next hop** IPv4 static route to the **LAN 2** network through Edge_Router.

Step 2: Configure IPv4 floating static routes to the internal LANs.

- On ISP1, configure a **directly connected** floating static route to LAN 1 through the ISP2 router. Use an administrative distance of **5**.
- On ISP1, configure a **directly connected** floating static route to LAN 2 through the ISP2 router. Use an administrative distance of **5**.

```
ISP1>
ISP1>ena
ISP1#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
ISP1(config)#ip route 192.168.10.16 255.255.255.240 10.10.10.2
ISP1(config)#ip route 192.168.11.32 255.255.255.224 10.10.10.2
ISP1(config)#ip route 192.168.10.16 255.255.255.240 g0/0 5
ISP1(config)#ip route 192.168.11.32 255.255.255.224 g0/0 5
ISP1(config)#
```

Copy Paste

กำหนดเส้นทางหลักแบบ next-hop ไปยังเครือข่าย LAN1 (192.168.10.16/28) และ LAN2 (192.168.11.32/27) ผ่าน Edge_Router (10.10.10.2) และกำหนด floating route สำรองผ่านอินเทอร์เฟซ G0/0 ไปทาง ISP2 โดยระบุ AD=5 เพื่อให้ ISP1 สามารถส่งข้อมูลกลับไปยัง LAN ภายในได้แม้เส้นทางหลักขาดการเชื่อมต่อ

Part 4: Configure IPv6 Static and Floating Static Routes to the Internal LANs.

Step 1: Configure IPv6 static routes to the internal LANs.

- On ISP1, configure a **next hop** IPv6 static route to the **LAN 1** network through Edge_Router.
- On ISP1, configure a **next hop** IPv6 static route to the **LAN 2** network through Edge_Router.

Step 2: Configure IPv6 floating static routes to the internal LANs.

- On ISP1, configure a **next hop** IPv6 floating static route to LAN 1 through the ISP2 router. Use an administrative distance of **5**.
- On ISP1, configure a **next hop** IPv6 floating static route to LAN 2 through the ISP2 router. Use an administrative distance of **5**.

If your configuration has been completed correctly, you should be able to ping the Web Server from the hosts on LAN 1 and LAN 2. In addition, if the primary route link is down, connectivity between the LAN hosts and the Web Server should still exist.

```
ISP1(config)#
ISP1(config)#
ISP1(config)#
ISP1(config)#ipv6 route 2001:db8:1:10::/64 2001:db8:a:1::2
ISP1(config)#ipv6 route 2001:db8:1:11::/64 2001:db8:a:1::2
ISP1(config)#ipv6 route 2001:db8:1:10::/64 2001:db8:f:f::2 5
ISP1(config)#ipv6 route 2001:db8:1:11::/64 2001:db8:f:f::2 5
ISP1(config)#
```

Copy

Paste

กำหนดเส้นทางหลักแบบ IPv6 next-hop ไปยังเครือข่าย LAN1 และ LAN2 ผ่าน Edge_Router (2001:db8:a:1::2) และกำหนด floating route สำรองผ่าน ISP2 (2001:db8:f:f::2) โดยระบุ AD=5 เพื่อเป็นเส้นทางสำรองในกรณีเส้นทางหลักล้ม เช่นเดียวกับที่ทำในฝั่ง IPv4

Part 5: Configure Host Routes

Users on the corporate network frequently access a server that is owned by an important customer. In this part of the activity, you will configure static host routes to the server. One route will be a floating static route to support the redundant ISP connections.

Step 1: Configure IPv4 host routes.

- On Edge Router, configure an IPv4 **directly connected** host route to the customer server.
- On Edger Router, configure an IPv4 **directly connected** floating host route to the customer server. Use an administrative distance of 5.

Step 2: Configure IPv6 host routes.

- On Edge Router, configure an IPv6 **next hop** host route to the customer server through the ISP1 router.
- On Edger Router, configure an IPv6 floating host route to the customer server through the ISP2 router. Use an administrative distance of 5.

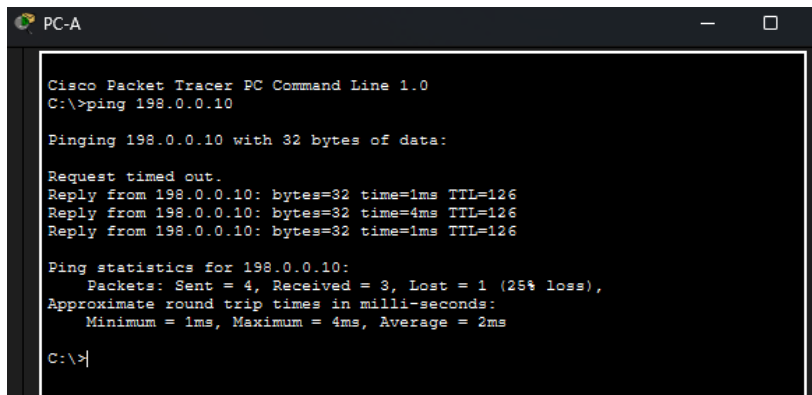
```
Edge_Router(config)#
Edge_Router(config)#
Edge_Router(config)#ip route 198.0.0.10 255.255.255.255 s0/0/0
Edge_Router(config)#ip route 198.0.0.10 255.255.255.255 s0/0/1 5
Edge_Router(config)#ipv6 route 2001:db8:f:f::10/128 2001:db8:a:1::1
Edge_Router(config)#ipv6 route 2001:db8:f:f::10/128 2001:db8:a:2::1 5
Edge_Router(config)#
```

กำหนด host route แบบ IPv4 ไปยัง Customer Server (198.0.0.10) โดยเส้นทางหลักผ่าน S0/0/0 (ไป ISP1) และเส้นทางสำรองผ่าน S0/0/1 (ไป ISP2) พร้อม AD=5 รวมถึงกำหนด host route แบบ IPv6 ไปยัง Customer Server (2001:db8:f:f::10/128) โดยเส้นทางหลักผ่าน ISP1 และเส้นทางสำรองผ่าน ISP2 (AD=5) เพื่อให้เครือข่ายภายในสามารถเข้าถึง Customer Server ได้อย่างต่อเนื่องแม้มีปัญหาที่เส้นทางใดเส้นทางหนึ่ง

ทดสอบผลลัพธ์

จาก PC-A

PC-A> ping 198.0.0.10



```
PC-A
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.0.0.10

Pinging 198.0.0.10 with 32 bytes of data:

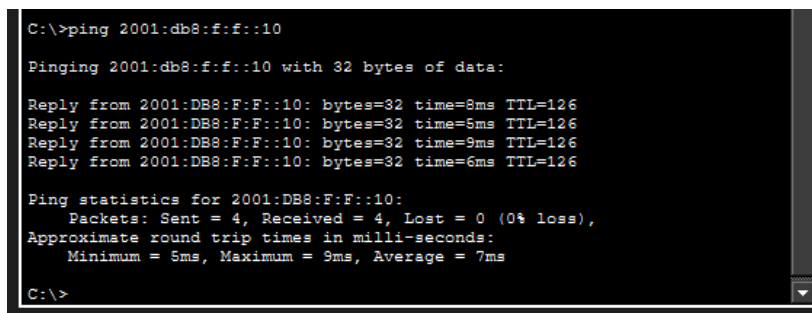
Request timed out.
Reply from 198.0.0.10: bytes=32 time=1ms TTL=126
Reply from 198.0.0.10: bytes=32 time=4ms TTL=126
Reply from 198.0.0.10: bytes=32 time=1ms TTL=126

Ping statistics for 198.0.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\>
```

มีการตอบกลับ

PC-A> ping 2001:db8:f:f::10



```
C:\>ping 2001:db8:f:f::10

Pinging 2001:db8:f:f::10 with 32 bytes of data:

Reply from 2001:DB8:F:F::10: bytes=32 time=8ms TTL=126
Reply from 2001:DB8:F:F::10: bytes=32 time=5ms TTL=126
Reply from 2001:DB8:F:F::10: bytes=32 time=9ms TTL=126
Reply from 2001:DB8:F:F::10: bytes=32 time=6ms TTL=126

Ping statistics for 2001:DB8:F:F::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 9ms, Average = 7ms

C:\>
```

มีการตอบกลับ

จาก PC-B

```
PC-B
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.0.0.10

Pinging 198.0.0.10 with 32 bytes of data:

Reply from 198.0.0.10: bytes=32 time=2ms TTL=126
Reply from 198.0.0.10: bytes=32 time=4ms TTL=126
Reply from 198.0.0.10: bytes=32 time=5ms TTL=126
Reply from 198.0.0.10: bytes=32 time=17ms TTL=126

Ping statistics for 198.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 17ms, Average = 7ms

C:\>ping 2001:db8:f:f::10

Pinging 2001:db8:f:f::10 with 32 bytes of data:

Reply from 2001:DB8:F:F::10: bytes=32 time=1ms TTL=126
Reply from 2001:DB8:F:F::10: bytes=32 time=8ms TTL=126
Reply from 2001:DB8:F:F::10: bytes=32 time=1ms TTL=126
Reply from 2001:DB8:F:F::10: bytes=32 time=1ms TTL=126

Ping statistics for 2001:DB8:F:F::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 2ms

C:\>
```

มีการตอบกลับ

ทดสอบเส้นทางกลับจาก ISP

ISP1#show ip route

```
ISP1
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

ISP1>
ISP1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.10.10.0/30 is directly connected, Serial0/0/0
L    10.10.10.1/32 is directly connected, Serial0/0/0
S    192.168.10.0/28 is subnetted, 1 subnets
     192.168.10.16/28 [1/0] via 10.10.10.2
S    192.168.11.0/27 is subnetted, 1 subnets
     192.168.11.32/27 [1/0] via 10.10.10.2
S    198.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.0.0.0/24 is directly connected, GigabitEthernet0/0
L    198.0.0.1/32 is directly connected, GigabitEthernet0/0

ISP1>
```

ISP1#show ipv6 route

```
ISP1
Physical Config CLI Attributes
192.168.11.0/27 is subnetted, 1 subnets
S   192.168.11.32/27 [1/0] via 10.10.10.2
C   198.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
L   198.0.0.0/24 is directly connected, GigabitEthernet0/0
L   198.0.0.1/32 is directly connected, GigabitEthernet0/0

ISP1>show ipv6 route
IPv6 Routing Table - 7 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S   2001:DB8:1:10::/64 [1/0]
    via 2001:DB8:A:1::2
S   2001:DB8:1:11::/64 [1/0]
    via 2001:DB8:A:1::2
C   2001:DB8:A:1::/64 [0/0]
    via Serial0/0/0, directly connected
L   2001:DB8:A:1::1/128 [0/0]
    via Serial0/0/0, receive
C   2001:DB8:F:F::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L   2001:DB8:F:F::1/128 [0/0]
    via GigabitEthernet0/0, receive
L   FF00::/8 [0/0]
    via Null0, receive
ISP1>
```

มีการตอบกลับทั้งหมด

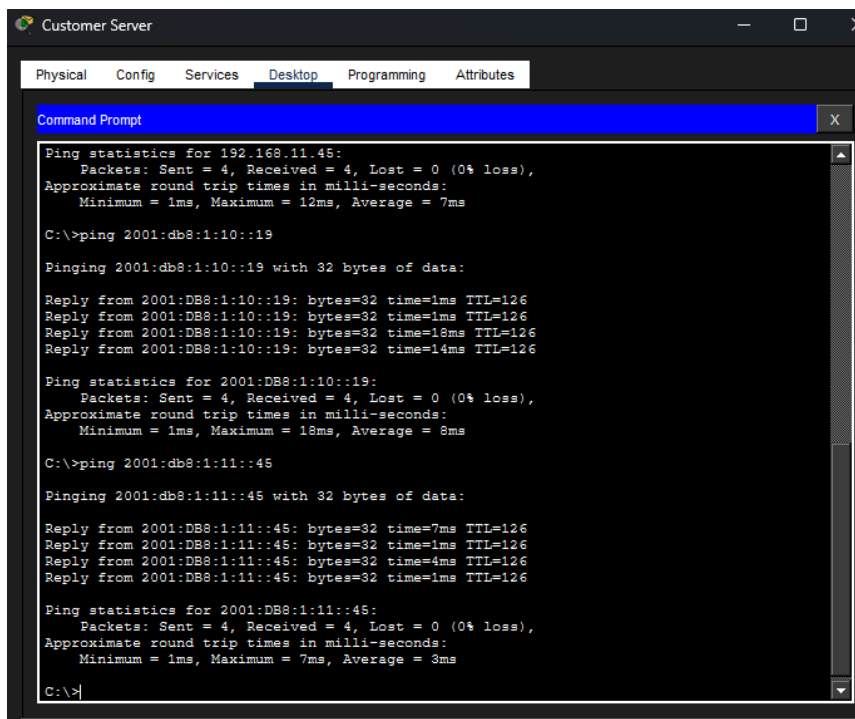
ทดสอบจาก Customer Server กลับมาที่ LAN

ping 192.168.10.19 (ไป PC-A)

ping 192.168.11.45 (ไป PC-B)

ping 2001:db8:1:10::19

ping 2001:db8:1:11::45



```
Customer Server
Physical Config Services Desktop Programming Attributes
Command Prompt
Ping statistics for 192.168.11.45:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 7ms

C:\>ping 2001:db8:1:10::19

Pinging 2001:db8:1:10::19 with 32 bytes of data:

Reply from 2001:DB8:1:10::19: bytes=32 time=1ms TTL=126
Reply from 2001:DB8:1:10::19: bytes=32 time=1ms TTL=126
Reply from 2001:DB8:1:10::19: bytes=32 time=18ms TTL=126
Reply from 2001:DB8:1:10::19: bytes=32 time=14ms TTL=126

Ping statistics for 2001:DB8:1:10::19:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 18ms, Average = 8ms

C:\>ping 2001:db8:1:11::45

Pinging 2001:db8:1:11::45 with 32 bytes of data:

Reply from 2001:DB8:1:11::45: bytes=32 time=7ms TTL=126
Reply from 2001:DB8:1:11::45: bytes=32 time=1ms TTL=126
Reply from 2001:DB8:1:11::45: bytes=32 time=4ms TTL=126
Reply from 2001:DB8:1:11::45: bytes=32 time=1ms TTL=126

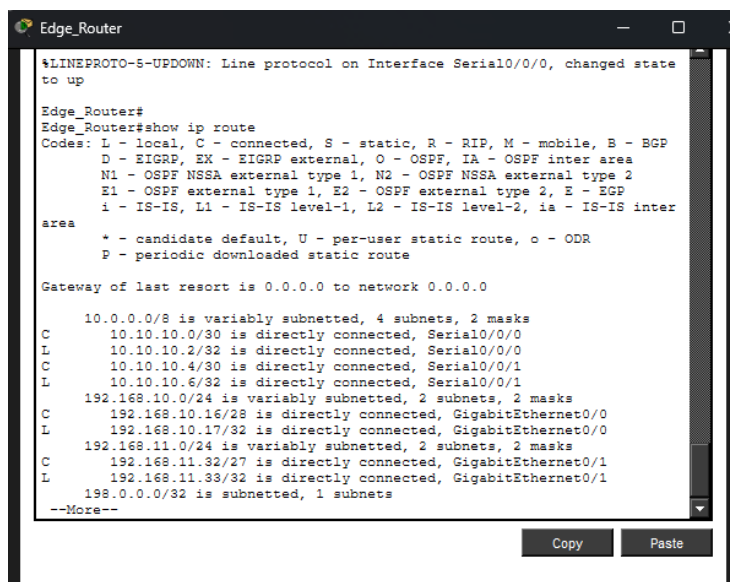
Ping statistics for 2001:DB8:1:11::45:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 7ms, Average = 3ms

C:\>
```

มีการตอบกลับทั้งหมด

เช็ค Routing Table ทุกเราเตอร์ให้ครบ

Edge_Router#show ip route



```
Edge_Router
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

Edge_Router#
Edge_Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

 10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.10.10.0/30 is directly connected, Serial0/0/0
L    10.10.10.2/32 is directly connected, Serial0/0/0
C    10.10.10.4/30 is directly connected, Serial0/0/1
L    10.10.10.6/32 is directly connected, Serial0/0/1
 192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.16/28 is directly connected, GigabitEthernet0/0
L    192.168.10.17/32 is directly connected, GigabitEthernet0/0
 192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.11.32/27 is directly connected, GigabitEthernet0/1
L    192.168.11.33/32 is directly connected, GigabitEthernet0/1
 198.0.0.0/32 is subnetted, 1 subnets
--More--

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```

Edge_Router#show ipv6 route

```
Edge_Router
Physical Config CLI Attributes

198.0.0.0/32 is subnetted, 1 subnets
S    198.0.0.10/32 is directly connected, Serial0/0/0
S*   0.0.0.0/0 is directly connected, Serial0/0/0

Edge_Router#show ipv6 route
IPv6 Routing Table - 11 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDP - ND Prefix, DCE - Destination, NDR - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S    ::0 [1/0]
    via 2001:DB8:A:1::1
C    2001:DB8:1:10::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L    2001:DB8:1:10::1/128 [0/0]
    via GigabitEthernet0/0, receive
C    2001:DB8:1:11::/64 [0/0]
    via GigabitEthernet0/1, directly connected
L    2001:DB8:1:11::1/128 [0/0]
    via GigabitEthernet0/1, receive
C    2001:DB8:A:1::/64 [0/0]
    via Serial0/0/0, directly connected
L    2001:DB8:A:1:2:2/128 [0/0]
    via Serial0/0/0, receive
C    2001:DB8:A:2::/64 [0/0]
    via Serial0/0/1, directly connected
L    2001:DB8:A:2:2:2/128 [0/0]
    via Serial0/0/1, receive
S    2001:DB8:F:F::10/128 [1/0]
```

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ISP1#show ip route

```
ISP1
C    2001:DB8:F:F::/64 [0/0]
    via GigabitEthernet0/0, directly connected
L    2001:DB8:F:F::1/128 [0/0]
    via GigabitEthernet0/0, receive
L    FF00::/8 [0/0]
    via Null0, receive
ISP1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.10.10.0/30 is directly connected, Serial0/0/0
L    10.10.10.1/32 is directly connected, Serial0/0/0
S    192.168.10.0/28 is subnetted, 1 subnets
    192.168.10.16/28 [1/0] via 10.10.10.2
S    192.168.11.0/27 is subnetted, 1 subnets
    192.168.11.32/27 [1/0] via 10.10.10.2
S    198.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.0.0.0/24 is directly connected, GigabitEthernet0/0
L    198.0.0.1/32 is directly connected, GigabitEthernet0/0

ISP1>
```

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ISP1#show ipv6 route

```
ISP1
192.168.11.32/27 [1/0] via 10.10.10.2
198.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.0.0.0/24 is directly connected, GigabitEthernet0/0
L    198.0.0.1/32 is directly connected, GigabitEthernet0/0

ISP1> show ipv6 route
IPv6 Routing Table - 7 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
S    2001:DB8:1:10::/64 [1/0]
     via 2001:DB8:A:1::2
S    2001:DB8:1:11::/64 [1/0]
     via 2001:DB8:A:1::2
C    2001:DB8:A:1::/64 [0/0]
     via Serial0/0/0, directly connected
L    2001:DB8:A:1::1/128 [0/0]
     via Serial0/0/0, receive
C    2001:DB8:F:F::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L    2001:DB8:F:F::1/128 [0/0]
     via GigabitEthernet0/0, receive
L    FF00::/8 [0/0]
     via Null0, receive
ISP1>
```

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Top

ISP2#show ip route

```
ISP2>
ISP2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.10.10.4/30 is directly connected, Serial0/0/1
L    10.10.10.5/32 is directly connected, Serial0/0/1
198.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.0.0.0/24 is directly connected, GigabitEthernet0/0
L    198.0.0.2/32 is directly connected, GigabitEthernet0/0

ISP2>
```

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ISP2#show ipv6 route

```

L      198.0.0.2/32 is directly connected, GigabitEthernet0/0
ISP2>show ipv6 route
IPv6 Routing Table - 5 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C  2001:DB8:A:2::/64 [0/0]
   via Serial0/0/1, directly connected
L  2001:DB8:A:2::1/128 [0/0]
   via Serial0/0/1, receive
C  2001:DB8:F:F::/64 [0/0]
   via GigabitEthernet0/0, directly connected
L  2001:DB8:F:F::2/128 [0/0]
   via GigabitEthernet0/0, receive
L  FF00::/8 [0/0]
   via Null0, receive
ISP2>

```

สรุป

Default route มีครบทั้ง 2 เส้นทาง ที่ Edge_Router

Floating route แสดงเป็น backup เมื่อลิงก์หลัก up อยู่

Host route ไปยัง Customer Server ปรากฏถูกต้อง